PIYUSH JHA

Python Developer | AI & LLM Engineering | Data Science @ IIT Madras

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CAREER SUMMARY

Python-focused AI developer with hands-on experience building scalable backend systems, REST APIs, and LLM-powered applications. Currently pursuing B.S. in Data Science at IIT Madras. Skilled in FastAPI/Flask, data pipelines, SQL databases, and integrating Large Language Models into real-time workflows. Interested in Generative AI, retrieval systems (RAG), and building research-grade AI tools.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++
Backend & APIs: FastAPI, Flask, REST APIs, Jinja2
AI/LLM: Prompt Engineering, RAG Pipelines, LangChain (learning), Embeddings, Scikit-learn
Data & Databases: MySQL, SQLite, SQLAlchemy, Data Modeling
Systems & Tools: Docker, Git, Linux, Jupyter
Cloud/Big Data (Learning): GCP, Spark, Kafka

PROJECT WORK

LLM Code Deployment & Analysis Pipeline | FastAPI, Python, LLMs | GitHub: Deployment Repo | Analysis Repo

- Designed and built a **FastAPI backend to integrate LLM inference** into real-time workflows.
- Designed REST endpoints for **prompt handling**, **model inference, and response caching.**
- Implemented modular preprocessing and response pipeline for scalable request handling.
- Reduced inference latency using asynchronous requests, batching, and response caching.
- Containerized with **Docker** for reproducible deployment.

Data Processing & Analytics Pipeline for Retail Business | Python (pandas, matplotlib) | GitHub: Repo

- Designed and implemented a data pipeline to process **87 days of operational data** across pricing, sales, employees, and expenses.
- Built automated **data cleaning & preprocessing pipeline** using pandas (NaNs, type coercion, normalization).
- Engineered SKU-level revenue, profit, and correlation analysis.
- Designed statistical summaries and visualizations to identify revenue drivers.
- Delivered actionable optimization strategies that improved profitability decisions.

Vehicle Telemetry ML Pipeline (End-to-End MLOps System) | Python, FastAPI, Scikit-learn, Pandas, Docker | GitHub: Repo

- Generated ***~3.9M rows of synthetic vehicle telemetry*** data and engineered 40+ time-series features.
- Trained ***RandomForest regression and classification models*** measured by R^2, MAE, and F1
- Built FastAPI inference service with batch predictions, health checks, and model version loading
- Reduced feature storage by ~80% using ***Apache Parquet*** for efficient columnar processing
- Containerized full pipeline (**data → features → train → API**) with ***Docker*** for reproducible deployment
- Deployed live demo service with REST endpoints and ***automated tests/CI***

Time Series Forecasting Pipeline | Python, XGBoost, ML | GitHub: Repo

- Built an end-to-end time-series forecasting pipeline for cinema audience prediction using multi-source booking and POS datasets
- Engineered time, seasonal, and autoregressive lag features for improved model performance
- Benchmarked baseline models *(Linear Regression, Random Forest)* and ***developed tuned XGBoost*** model with ***RMSE ~24.6 & R^2~0.45***
- **Ranked 826/2632 (top ~31%) on Kaggle leaderboard; score 0.334** › surpassing cutoff score for competition success

EDUCATION

BACHELOR'S DEGREE | B.S. Data Science and Applications, IIT Madras | **CGPA: 8.07/10** Ongoing
BACHELOR'S DEGREE | Computer Application, Maulana Abul Kalam Azad University of Technology, West Bengal | **CGPA: 9.04/10** 2022

RESEARCH & TECHNICAL INTERESTS

Generative AI • Retrieval-Augmented Generation (RAG) • LLM-powered Systems • MLOps • Big Data Engineering (GCP, Spark, Kafka)